DPUTRFSNDX

1. Inputs:

ignoreTRFid	TRF-ID which is not simulated default: 0 (ego-vehicle)
GlobalVolume	Global volume of the sound sources
EngineVolume	volume of the engines sound sources
WindRollVolume	volume of the sound sources for wind and roll noise
MiscVolume	volume of the misc sound sources
ReferenceDistance	Distance [m] before a sound source softens (according to the OpenAL sound model) default: 5
RolloffFactor	Factor for the above mentioned softening. Values < 1 lead to slower softening, for values > 1 it is faster. default: 1

2. Parameters:

NSources	Maximum number of simulated vehicles (A vehicle may have more than one source!) [Maximum number of available sound sources depends on the hardware] default: 10
Radius	Vehicle sounds are simulated within this radius [m] (If this number is higher than *NSources* simulation of the vehicles closest to the listener is done.) default: 100
BufferCount	Number of buffers for each sound source default: 10
BufferLengthMS	Length of each buffer [ms] default: 16

3. Outputs:

NActiveSources	Number of sound sources currently in use
ComputeTimeMS	Computation time of the DPU [ms] for debug purposes

4. Functional principle:

DPUTRFSNDX was developed to simulate the sound of surrounding vehicles. For every of the *NSources* sources multiple sound sources are initialized. Type and number may be specified in the configuration.

For driving noises VSTi-plugins are used [The program SynthMaker was used to create them.] which are available as a DLL. Any VSTi-plugin is applicable for the driving noise simulation as long as it has at least inputs for „RPM“, „Last“ and „v-1“. These inputs are calculated to match the vehicle to be simulated but they also can be overriden in the configuration section.

The current implementation uses different parameter sets of the VSTi-plugin depending on the vehicle-ID.

Note:

Maximum number of sound sources for a Realtek-Onboard-Soundcard is about 32. The Creative Soundblaster X-Fi can handle up to 128 sources. 256 Quellen, allerdings wird der Prozessor dadurch sehr stark belastet.

In any case one source is reserved for EAX as well as two other blocked sources for stereo playback (Standard sound simulation SILAB).

5. Example for a driving noise configuration:

```
SILAB ASM
{
	DrivingNoise dn1
	{
		TRFType = {1};
		VSTi = "pkw_engine";
		VSTiParams v1
		{
			Volume = 0.9;
			MotorGain = 0.8;
			NoiseGain = 1.0;
			EQ-On = 1.0;
			EQ-Freq1 = 0.34;  # [22.6274..17818] Hz
			EQ-Q1    = 0.12;  # [0.1..8]
			EQ-Gain1 = 0.56;  # [-40..40]
			EQ-Freq2 = 0.20;
			EQ-Q2    = 0.01;
			EQ-Gain2 = 0.25;
			EQ-Freq3 = 0.22;
			EQ-Q3    = 0.16;
			EQ-Gain3 = 0.50;
			EQ-Freq4 = 0.01;
			EQ-Q4    = 0.50;
			EQ-Gain4 = 0.55;
			Waveshaper-X1 = 0.3; # [-1..1]
			Waveshaper-X2 = 0.7;
			Waveshaper-X3 = 0.6;
		};
		VSTiParams v2
```

```
			{
				Volume = 0.8;
				MotorGain = 0.7;
				NoiseGain = 1.0;
				EQ-On = 0.0;
			};
			VSTiParams v3
			{
				# empty set uses default values (preset)
			};
		};
		DrivingNoise dn2
		{
			TRFType = {2};
			VSTi = "lkw_engine";
			VSTiParams v1
			{
				Volume = 0.7;
			};
/* set deactivated
* preset values will be used
			VSTiParams v2
			{
				Volume = 0.7;
			};
*/
			VSTiParams v3
			{
				Volume = 0.8;
			};
		};

};
```

The configuration consists of the following parts:

SILAB ASM marks this section as configuration for DPUTRFSNDX.

DrivingNoise declares a sound source which is designated for vehicle sounds done by a VSTi-plugin. This section‘s parameter *VSTi* is a mandatory and should be the filename of a valid VSTi-plugin as is the *TRFType* list which defines which vehicles can be represented by this VSTI-plugin (0 = ego-vehicle, 1 = passenger car, 2 = truck).

Additional parameters can be set. The above example uses the *VSTiParams* v1 to v3. During initialization the parameters are checked against the plugin. If a invalid parameter is detected an error message is displayed on loading time. In this example the VSTi-plugin *pkw_engine.dll* has parameters for a global volume (Volume), as well as a seperate volume control for engine (MotorGain) and wind (NoiseGain). Furthermore a four-band-equalizer and a wave-shaper are available to manipulate the sound even more.

Any values not set in the configuration will keep their preset value. The first preset of the VSTi plugin that can be found is used for this. Therefor one might have exactly one preset with reasonable values.

Should it happen that in one simulation step are more vehicles within the radius than sources available, a message is displayed once. When stoping the simulation another message is displayed to inform how many simulations steps have been done without sufficent sources.

As an alternative to *DrivingNoise* the sound simulation file from Allersmeier may be used. The configuration section then reads:

```
SILAB ASM
{
	EngineSim es1
	{
		SIMfile = "BMW520E34V17.SIM";
	};
};
```

Only the engine of the surrounding vehicles may be simulated then. Depending on the vehicle-ID the rounds per minute are varied to a certain extend. *EngineSim* cannot distinguish between different vehicle types.